Date of Assessment: October 13, 2026
Location: Bolo Town, Nimba County
Sample Matrix: Water
Analysis Start Date: October 15, 2026

Table 1: Surface Water Quality Sample Information

Sample Location	Sample ID	GPS Coordinates		Sampling Time	Description
		Latitude	Longitude		
Wenee Downstream	WE-JC-SW-001	0542501	0834879	2:08 PM	—
Wenee mid-point	WE-JC-SW-002	0542460	0834664	2:21 PM	—
Wenee Upstream (confluence with Saymenee)	WE-SW-JC-003	0542353	0834524	2:36 PM	—
Saymenee mid-point	SM-SW-JC-001	0542304	0834561	2:45 PM	—
Alphonso farm	AF-SW-JC-001	7.547899	-8.616177	2:21 PM	—
Neneyee Upstream (northern sediment control station)	NEN-SW-JC-001	7.537928	-8.614119	4:24 PM	—
Wehyee Downstream	W-SW-VA-001	7.56771	-8.60949	2:58 PM	Downstream surface water sample collected along the Wehyee stream within a densely vegetated channel. Water was brownish to reddish-brown and visibly turbid, with suspended particles and sediment apparent in the flowing water. No visible sign of mine-runoff pollution was observed on the nearby farmland.
Wehyee Mid-point	W-SW-VA-002	7.56695	-8.61227	3:16 PM	Mid-point surface water sample collected along the Wehyee stream to assess spatial variation in sediment load. The stream was turbid and reddish-brown, with uneven banks showing exposed soil, roots and localised erosion along the channel.
Wehyee Upstream	W-SW-VA-003	7.56724	-8.6128	3:21 PM	Upstream surface water sample collected on the Wehyee stream above the mid-point and downstream stations. Water remained discoloured and turbid under dense tree, shrub and fern cover; disturbed bank sections with exposed soils were noted as possible local sediment sources.
Pompleh Downstream	POM-SW-VA-001	7.56793	-8.6094	3:03 PM	Downstream surface water sample collected along the Pompleh stream within dense vegetation. Brownish, visibly turbid water with suspended sediment was observed; no visible sign of mine-runoff pollution was noted on the adjacent farmland.
Pompleh Mid-point	POM-SW-VA-002	7.56822	-8.60947	3:07 PM	Mid-point surface water sample collected along the Pompleh stream. The channel was turbid and reddish-brown, flowing through abundant ground vegetation with uneven, partially eroded streambanks.

Sample Location	Sample ID	GPS Coordinates		Sampling Time	Description
		Latitude	Longitude		
Pompleh Downstream	POM-SW-VA-003	7.56882	-8.60976	3:35 PM	Surface water sample collected further along the Pompleh stream to evaluate the combined sediment load. Water was discoloured and turbid, with suspended particles consistent with soil erosion and sediment runoff into the stream.
Yeewa Downstream	YEEWA-DS	542444	834881	2:09 PM	Downstream surface water sample collected from the Yeewa main stream, a natural water body, below the inflow of the Yeewa Korweny 1 and 2 tributaries. Iron ore sediment was observed along the waterway and turbidity was visibly higher than at Yeewa Korweny 1.
Yeewa Mid-Stream	YEEWA-MS	542314	834915	2:16 PM	Mid-stream surface water sample collected along the Yeewa main stream to assess spatial sediment migration. Reddish iron ore sediment deposits were observed along the channel, with elevated turbidity.
Yeewa Up-Stream	YEEWA-US	542066	835147	2:32 PM	Upstream surface water sample collected on the Yeewa main stream above the tributary junctions. Iron ore sediment was still present along the waterway at this point, indicating sediment loading from further upstream in the Mount Yewlliton drainage.
Yeewa Korweny Patel Downstream	YWK2-ES-SW-001	0541616	0835024	3:15 PM	Downstream surface water sample collected from Yeewa Korweny 2 (Patel), a tributary of the Yeewa main stream, just above its junction with the main watercourse. Substantial sediment deposits were observed and turbidity was higher than at Yeewa Korweny 1.
Yeewa Korweny Patel Midstream	YWK2-ES-SW-002	0541600	0834909	3:18 PM	Midstream surface water sample collected along Yeewa Korweny 2 (Patel) within a deep, densely forested valley. A geo-fencing sign reading “KEEP AWAY – 500 Meters Safety Buffer Zone” was observed on this tributary, in proximity to ArcelorMittal’s haulage route.
Yeewa Korweny Patel Upstream	YWK2-ES-SW-003	0541550	0834786	3:21 PM	Upstream surface water sample collected on Yeewa Korweny 2 (Patel) towards its source in the Mount Yewlliton area, in proximity to mining activities in the Mount Yewlliton and Mount Gangro areas. Sediment deposits were observed in the channel.
Yeewa korweny Dor Downstream	YWK1-ES-SW-001	0541697	0835051	3:35 PM	Downstream surface water sample collected from Yeewa Korweny 1 (Dor), the tributary on the opposite side of the Yeewa main stream from Yeewa Korweny 2, near its junction with the main watercourse. Sediment deposits were present but turbidity was visibly lower than at Yeewa and Yeewa Korweny 2.
Yeewa Korweny Dor Midstream	YWK1-ES-SW-002	0541644	0835167	3:44 PM	Midstream surface water sample collected along Yeewa Korweny 1 (Dor) as it flows through a deep valley under dense forest cover. Sediment deposits were observed, with lower turbidity than the Yeewa Korweny 2 branch.
Yeewa korweny Dor Midstream	YWK1-ES-SW-003	0541607	0835271	3:59 PM	Upper surface water sample collected on Yeewa Korweny 1 (Dor) towards its source in the Mount Yewlliton area, located in proximity to ArcelorMittal’s haulage route and the Mount Yewlliton / Mount Gangro mining areas. Natural water body with sediment deposits in the channel.

Note: Descriptions for the shaded rows are derived from the groups' field notes – "ES Group – Physical Observations at Following Sampling Points" (Emmanuel Suah group; YWK2 = Yeewa Korweny 2 (Patel), YWK1 = Yeewa Korweny 1 (Dor)) and "Team Varney Armah (Pompleh & Wehyee)" (VA group). Descriptions for the JC group points are to be supplied by that group.

PHYSICAL OBSERVATIONS AT SAMPLING POINTS – ES GROUP (YEEWA, YEEWA KORWENY 1 & 2)

Sampling was followed by a physical inspection of the Yeewa main stream and its Yeewa Korweny 1 (Dor) and Yeewa Korweny 2 (Patel) tributaries. The following preliminary observations were recorded by the ES group:

- An agricultural farm was observed to have been inundated and covered by sediments originating from the mining area. This may affect the overall productivity of the farmland.
- The presence of iron ore sediment was observed along the waterway during the sampling.
- The turbidity levels at Yeewa and Yeewa Korweny 2 were higher than those at Yeewa Korweny 1.
- A geo-fencing sign was observed at Yeewa Korweny 2 bearing the inscription: "KEEP AWAY – 500 Meters Safety Buffer Zone".
- All the sampling points were located within natural water bodies.
- Yeewa Korweny 1 and Yeewa Korweny 2 are tributary branches of the Yeewa main stream, situated on opposite sides of the main watercourse. Both originate in the Mount Yewlliton area and flow through deep valleys characterised by dense forest cover. The tributaries were also observed to contain substantial sediment deposits. Their locations are in proximity to ArcelorMittal's haulage route and to mining activities in the Mount Yewlliton and Mount Gangro areas. The proximity of mining and haulage activities, together with the observed sedimentation, warrants further investigation into possible sediment transport pathways. Nevertheless, the field observations alone do not establish that ArcelorMittal's activities caused the sedimentation or any chemical pollution.

PHYSICAL OBSERVATIONS AT SAMPLING POINTS – VA GROUP (POMPLEH & WEHYEE)

During the field visit on 13 October 2026, the Varney Armah (VA) team conducted surface water sampling along the Wehyee and Pompleh streams in Nimba County. The following preliminary observations were recorded:

- The stream water was brownish to reddish-brown and visibly turbid, with suspended particles and sediment apparent in the flowing water.
- Both streams flow through a densely vegetated environment characterised by abundant trees, shrubs, ferns and other ground vegetation.
- The streambanks were uneven, with exposed soil, roots and areas of erosion visible along the channel.
- The presence of suspended sediment and the water's discoloration may indicate soil erosion and sediment runoff into the stream, potentially influenced by rainfall and surface-water movement. However, the specific cause of the discoloration cannot be confirmed through visual observation alone.
- The surrounding vegetation provides natural streambank cover, although localised exposed soils and disturbed bank sections may contribute to sediment entering the water.
- Around the collection points along the Pompleh and Wehyee streams, there were no visible signs of possible environmental pollution due to mine runoff on the nearby farmlands.
- Water samples were collected for subsequent analysis to assess the streams' water quality and determine whether their physical and chemical characteristics meet the applicable environmental standards.

Photographic documentation – VA group sampling points

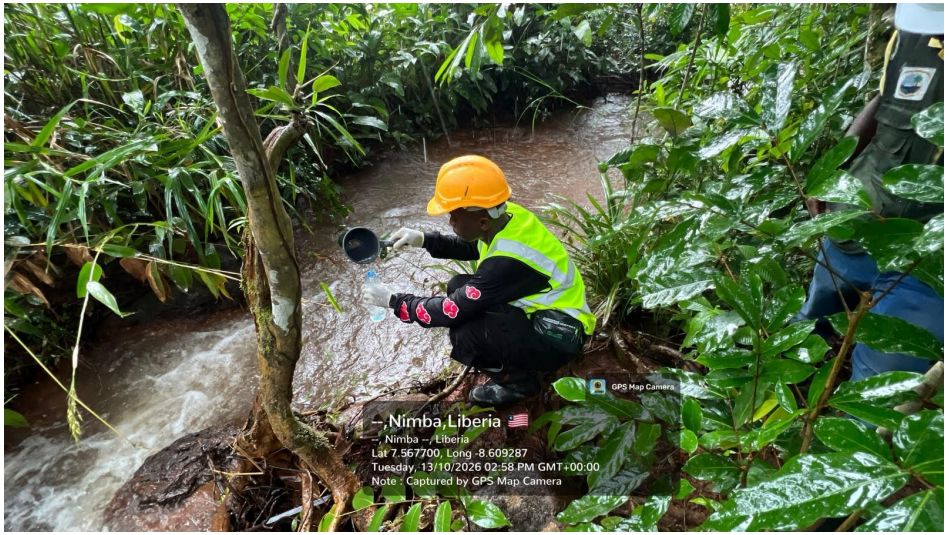


Plate 1: W-SW-VA-001 – Wehyee Downstream (7.5677, -8.6093; 2:58 PM). Reddish-brown, turbid flow.

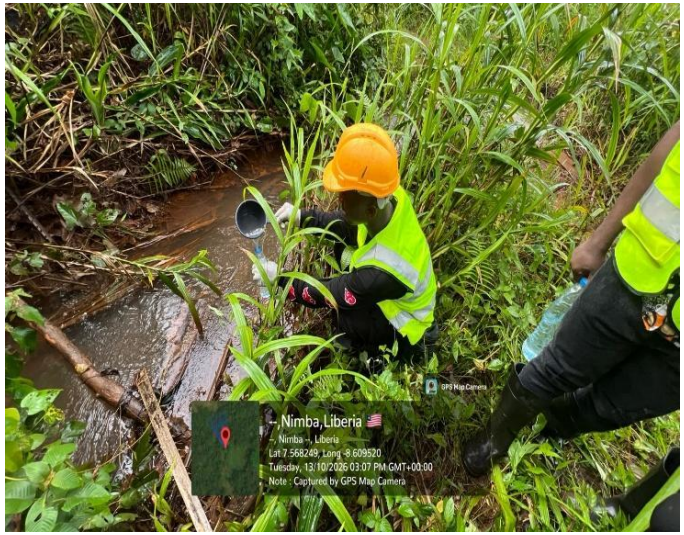


Plate 2: POM-SW-VA-002 – Pompleh Mid-point (7.5682, -8.6095; 3:07 PM). Sampling under dense ground cover.

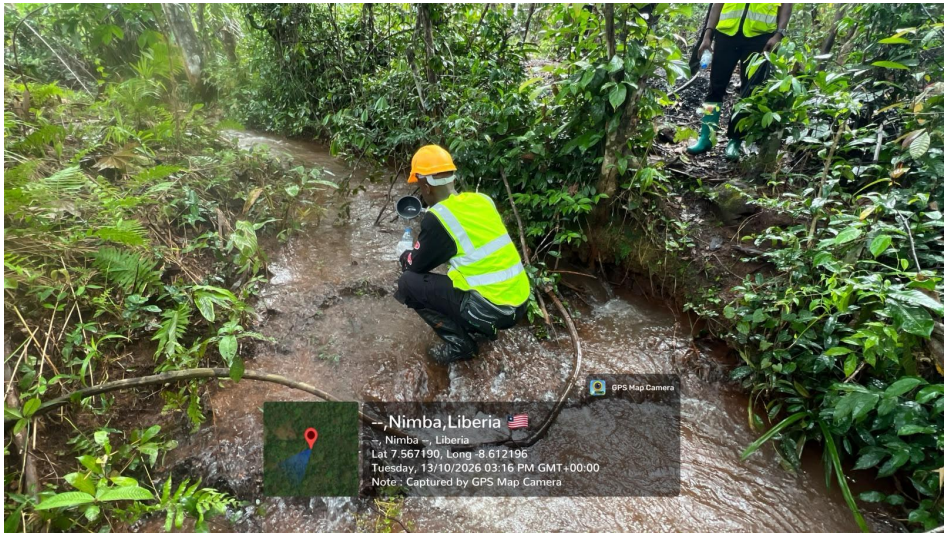


Plate 3: W-SW-VA-002 – Wehyee Mid-point (7.5672, -8.6122; 3:16 PM). Suspended sediment in channel.



Plate 4: W-SW-VA-003 – Wehyee Upstream (7.5672, -8.6129; 3:21 PM). Uneven banks with exposed soil.

Table 2: Surface Water Quality Result

Sample Code	Sample Parameter	LWQS Class III	Instrumentation
	Total Suspended Solid (TSS)		
YEEWA-DS	532		
YEEWA-MS	523		
YEEWA-US	628		
YWK2-ES-SW-001	1858		
YWK2-ES-SW-002	2,400		
YWK2-ES-SW-003	769		
YWK1-ES-SW-001	259		
YWK1-ES-SW-002	246		
YWK1-ES-SW-003	188		
W-SW-VA-001	24		
W-SW-VA-002	24	**≤50.0**	**Colorimeter**
W-SW-VA-003	20		
POM-SW-VA-001	11		
POM-SW-VA-002	24		
POM-SW-VA-003	164		
WE-JC-SW-001	102		
WE-JC-SW-002	118		
WE-SW-JC-003	214		
SM-SW-JC-001	84		
AF-SW-JC-00	1067		
NEN-SW-JC-001	92		